Marwadi University
Marwadi Chandarana Group

NAAC A+

MEDICINAL CHEMISTRY– III
(13PH0601)
UNIT-2
Anti-Malarials

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Topics

Antibiotics: Historical background, Nomenclature, Stereochemistry, Structure-activity relationship, Chemical degradation classification, and important products of the following classes.

Macrolide: Erythromycin, Clarithromycin, Azithromycin.

Miscellaneous: Chloramphenicol*, Clindamycin

Antimalarials: Etiology of malaria.

Prodrugs: Basic concepts and application of prodrugs design.

Quinolines: SAR, Quinine sulphate, Chloroquine*, Amodiaquine, Primaquine phosphate, Pamaquine*, Quinacrine hydrochloride, Mefloquine.

Biguanides and dihydro triazines: Cycloguanil pamoate, Proguanil.

Miscellaneous: Pyrimethamine, Artesunete, Artemether, Atovoquone.

Introduction :

➤ Malaria is a life-threatening disease caused by parasites from the Plasmodium genus, transmitted to humans through the bites of infected **Anopheles mosquitoes**.

➤ Malaria remains a significant global health issue, particularly in **tropical and subtropical** regions, affecting millions of people annually

.➤ Malaria is primarily transmitted through the bite of an infected female Anopheles mosquito.

Other modes of transmission include:

•Blood transfusions, Organ transplants, Sharing needles or syringes, From a pregnant mother to her unborn child (congenital malaria).

Etiology of Malaria :

Malaria is a parasitic disease caused by protozoan parasites of the genus Plasmodium, transmitted through the bite of infected Anopheles mosquitoes.

There are four species that infect humans:

1. **Plasmodium falciparum**: The most **dangerous** and lethal form, causing the majority of malaria-related deaths, particularly in sub-Saharan Africa.
2. **Plasmodium vivax**: Causes a **milder** form of malaria but can remain dormant in the liver, leading to declines.
3. **Plasmodium ovale**.
4. **Plasmodium malariae.**

Life Cycle

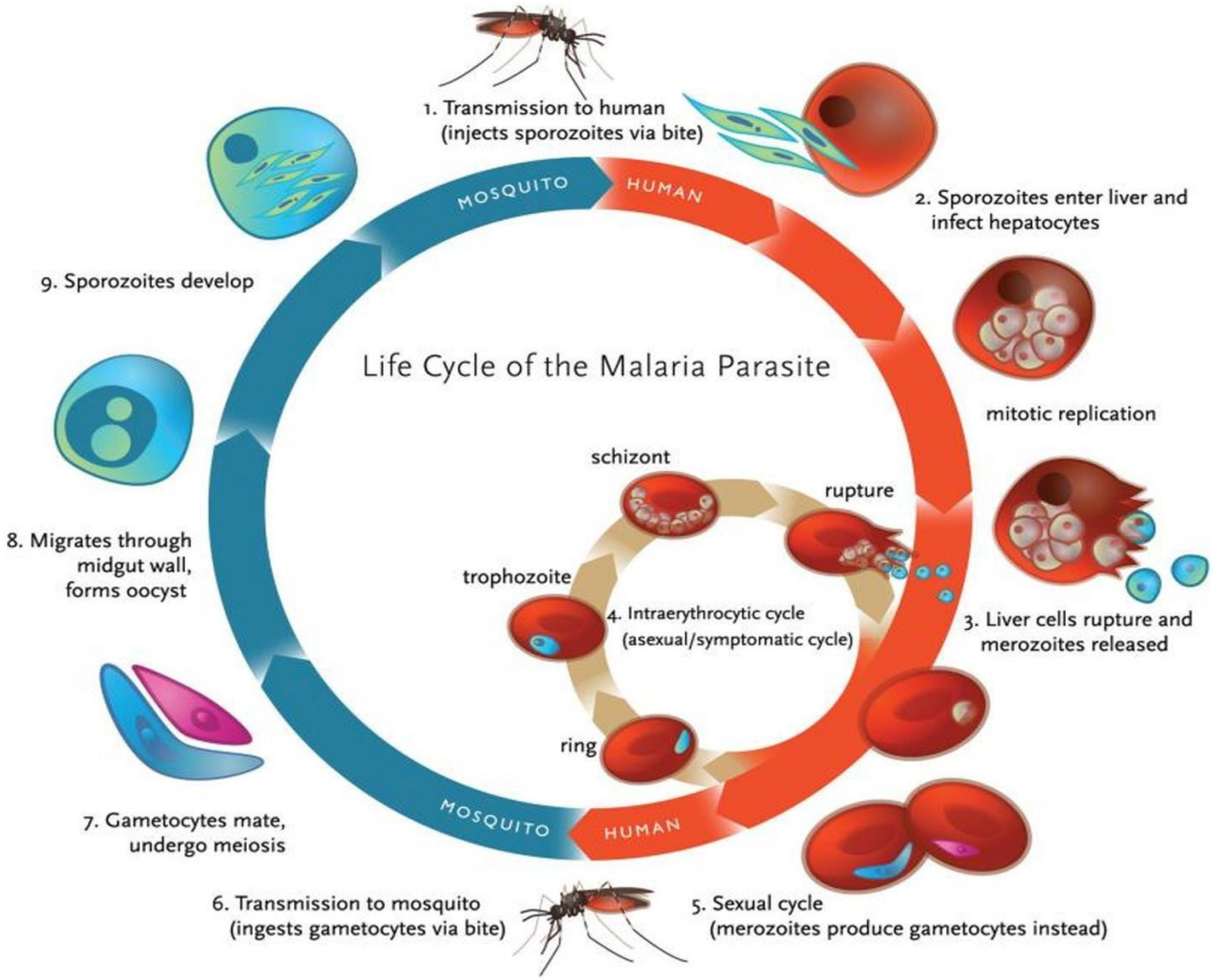
The life cycle of Plasmodium involves two hosts: mosquitoes and humans.

1.Mosquito Stage: The parasite develops in the Gametocyte mosquito's gut and salivary glands.

2.Liver Stage (Human): Upon biting, infected mosquito injects sporozoites into the human bloodstream, which travel to the liver, attack liver cells, and multiply. This stage is **asymptomatic.**

3.Blood Stage (Human): After maturing in the liver, parasites (merozoites) are released into the bloodstream, where they invade red blood cells, multiply, and cause the clinical symptoms of malaria.

4.Transmission Back to Mosquito: Some parasites develop into sexual forms (gametocytes) that, when ingested by another mosquito, continue the cycle



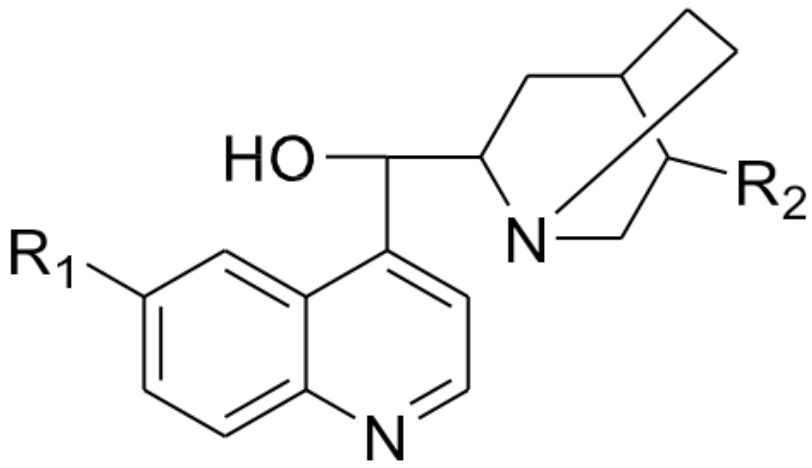
Classification of Antimalarial drugs

Classification	Name
Chinoca Alkaloids	Quinine
4 Aminoquionoline	Chloroquine,Hydroxychloroquine
8 Aminoquionoline	Primaquine
Biguanides	Proguanil
Diaminopyridines	Pyrimethamine
Quinoline methanol	Mefloquine
Phenantherene i. Artemisinin derivative ii. Acridine	Halofantherine i. Arthesunate, Artemeter, Arteether, ii. Mepacrine Quinacarine
Misellaneous	Sulfonamides, tetracycling,atavaquone

CLASSIFICATION

1.] Quinolines:

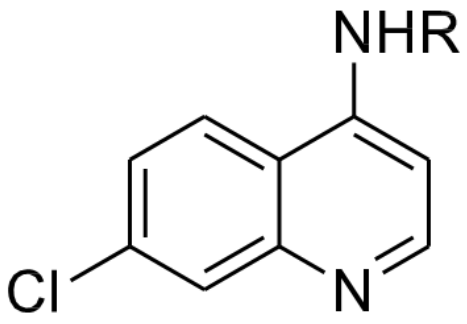
(a) **Chinchona alkaloids** : Quinine, Qunidine, Cinchonine, Cinchonidine

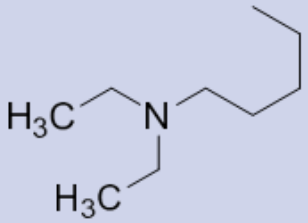
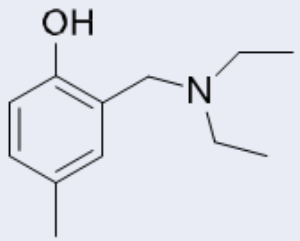


Compound	R1	R2	Isomerism
Quinine	-OCH3	-CH=CH2	Laevo (-)
Qunidine	-OCH3	-CH=CH2	Dextro (+)
Cinchonine	-H	-CH=CH2	Dextro (+)
Cinchonidine	-H	-CH=CH2	Laevo (-)

These are derivatives of **4-quinolinemethanol** which is linked with a substituted **quinuclidine** moiety.

(b) **4-Aminoquinolines:** Chloroquine, Amodiaquine



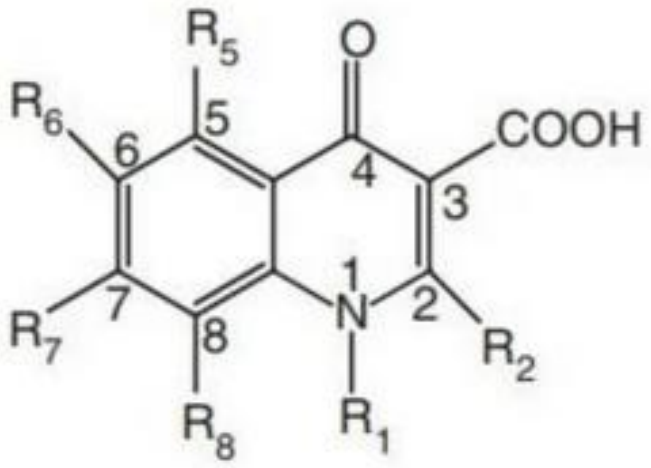
COUMPOUND	R
Chloroquine	
Amodiaquine	

The early quinolines were associated with a high toxicity profile and low therapeutic index.

To overcome this situation, 4-aminoquinolines were investigated

SAR of Quinolones

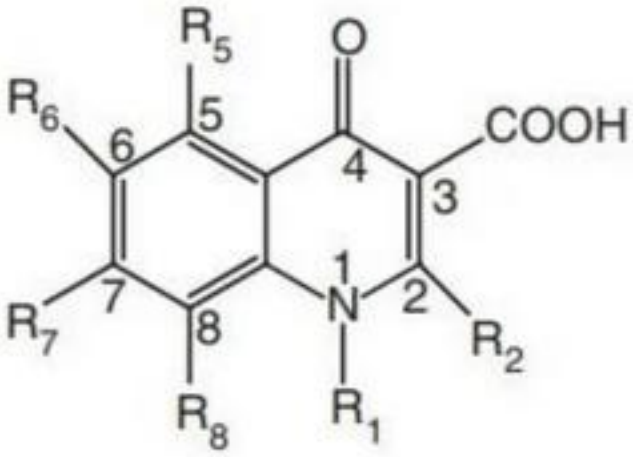
Quinolones are a class of synthetic antimicrobial agents that interfere with DNA synthesis. The SAR of quinolones in antimalarial drug development typically revolves around modifications to their core structure to improve efficacy, spectrum of activity, and pharmacokinetic properties.



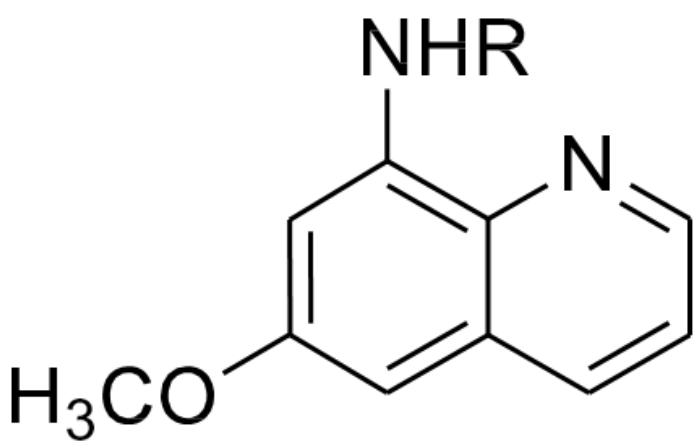
1. **Core Structure**: Quinolones possess a bicyclic ring structure that typically includes a quinoline or quinolone nucleus.
2. **Substitutions on the Quinolone Core**

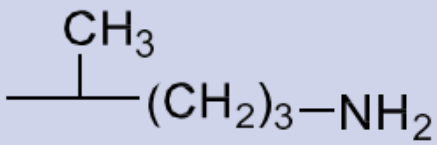
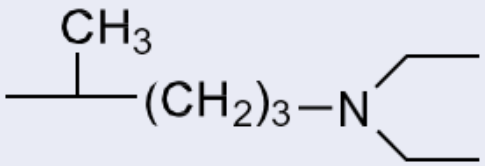
- **C-3 and C-4 Positions:** The ketone and carboxylic acid groups at these positions are critical for DNA gyrase inhibition and are conserved across quinolones.

- **N-1 Substituent:** Changes to this position, such as alkyl or aryl or isopropyl groups, can influence the potency, oral absorption, and spectrum of activity against Gram (-) bacteria.
- **C-7 Substituent:** Modifications here, like piperazine or other bulky groups, enhance the ability to penetrate microbial cells and can improve antimalarial activity.
- **C-8 Substituent**: Enhancing the pharmacokinetic properties, such as increasing half- life and reducing toxicity, can be achieved by altering the substituent at this position.(-NH_2 or –CH_3)
- Overall, these structural modifications optimize the pharmacodynamics and pharmacokinetics, making quinolones effective against various stages of the Plasmodium lifecycle.



(c) 8-Aminoquinolines: Primaquine, Pamaquine



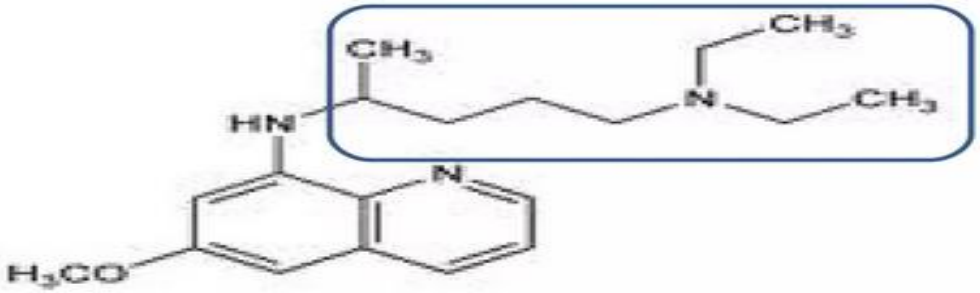
Compound	R
Primaquine	
Pamaquine	

Drug in this group have Amino group at position – 8 of quinoline ring.

Such drugs have OCH_3 group at position 7

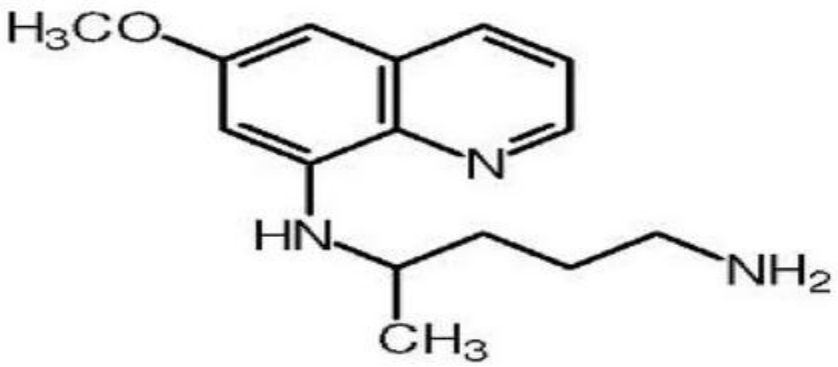
This molecule has antimalarial activity but when side chain is introduced at amino group antimalarial activity is intensified e.g pamaquine

It causes hemolysis of RBCs



Diethyl amino pentyl side chain

- It contains tertiary amino group and when it is converted into primary amino group the compound is called primaquine, which is
  - Less toxic
  - Well tolerated

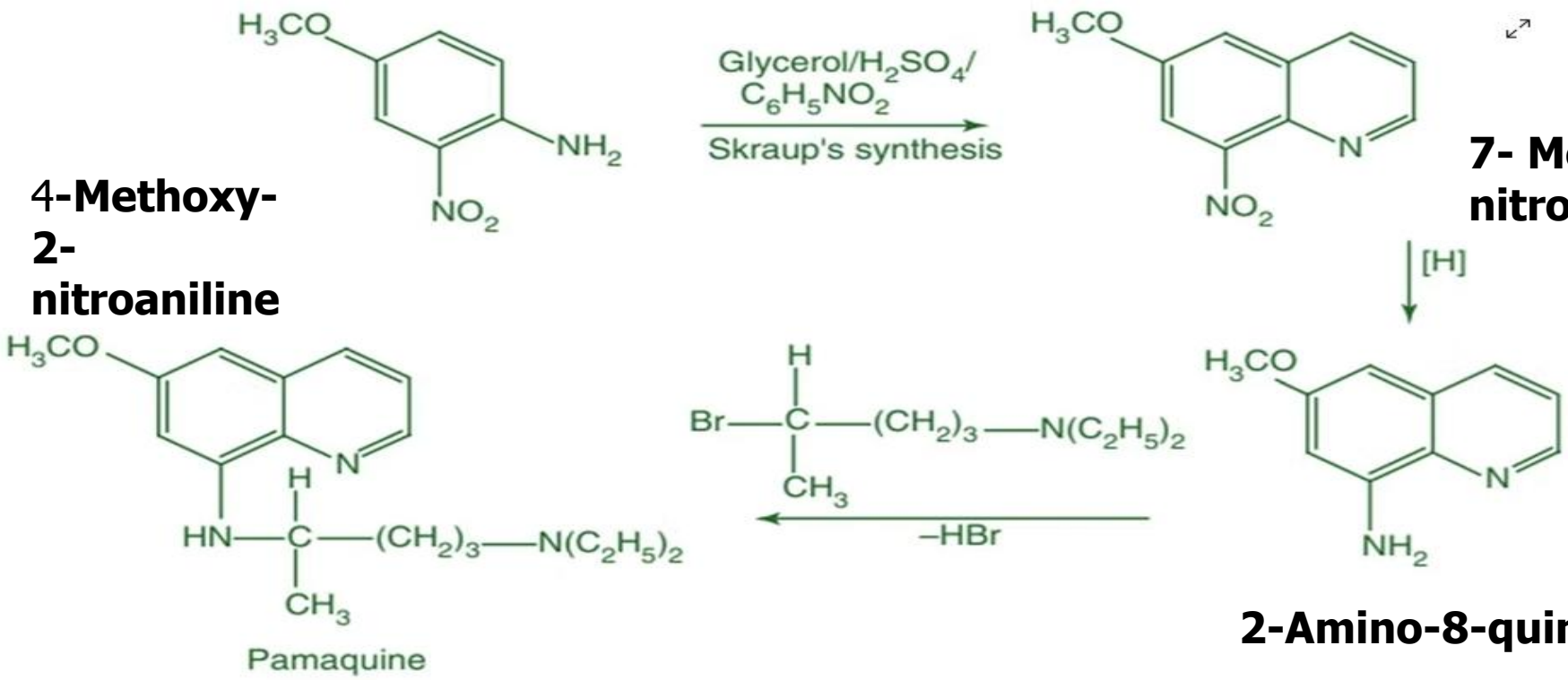


  - It is the most commonly used agent in this group in the treatment of malaria

- OCH_3 is not necessary for antimalarial activity but when replaced by OC_2H_5 the compound became
  - less active
  - Toxic in nature
- OCH_3 when replaced by CH_3 the compound become inactive
- Introduction of halogens increases toxicity
- Presence of quinoline ring is necessary for antimalarial activity. When pyridine ring is converted to piperidine (saturated) the compound became inactive

Pamaquine:

Synthesis of Pamaquine



4-Methoxy-2-nitroaniline

7- Methoxy-8-nitroquinoline

2-Amino-8-quinolinol

MOA of Pamaquine

Drug administration

↓

Pamaquine enters infected liver cells and RBCs

↓

Metabolic activation → formation of **reactive metabolites**

↓

Generation of **Reactive Oxygen Species (ROS)**

↓

Oxidative damage to mitochondria, enzymes, and DNA

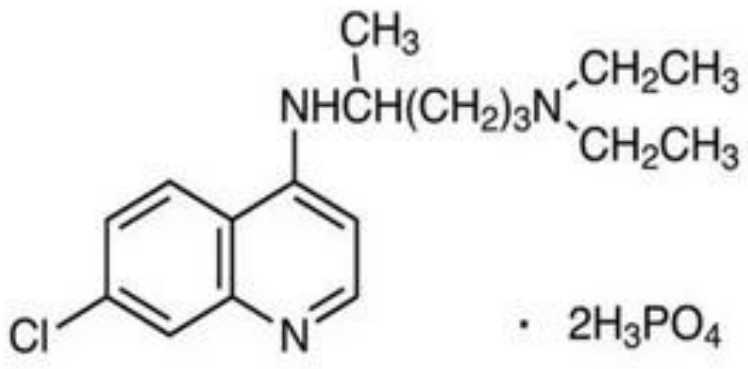
↓

Disruption of parasite metabolism

↓

Death of malaria parasite

Chloroquine Phosphate



Mechanism of Action

Drug administration

↓

Drug enters infected red blood cells

↓

Accumulation in **parasite food vacuole**

↓

Hemoglobin digestion releases **toxic free heme**

↓

inhibits heme polymerization (heme → hemozoin)

↓

Accumulation of toxic heme in parasite

↓

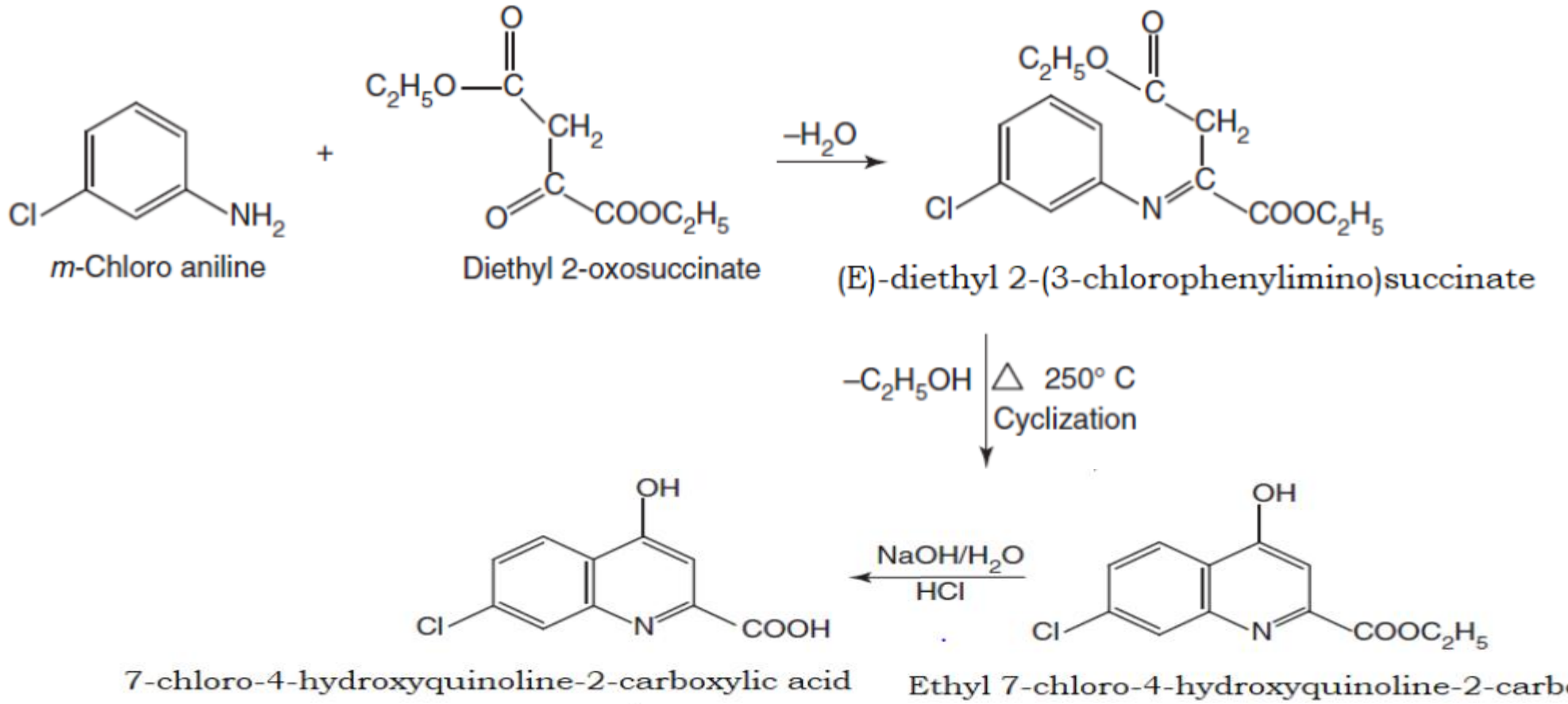
Membrane damage and oxidative stress

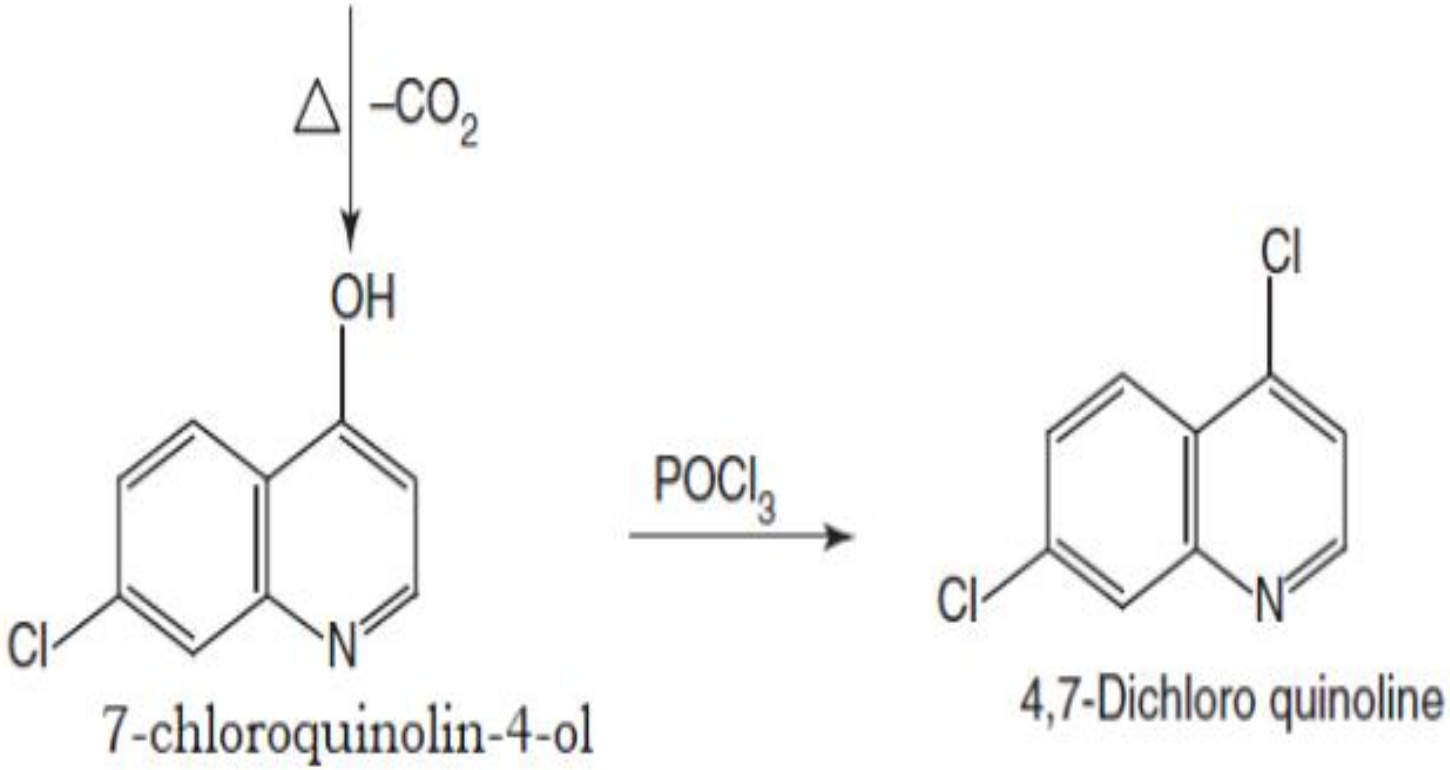
↓

Death of malaria parasite

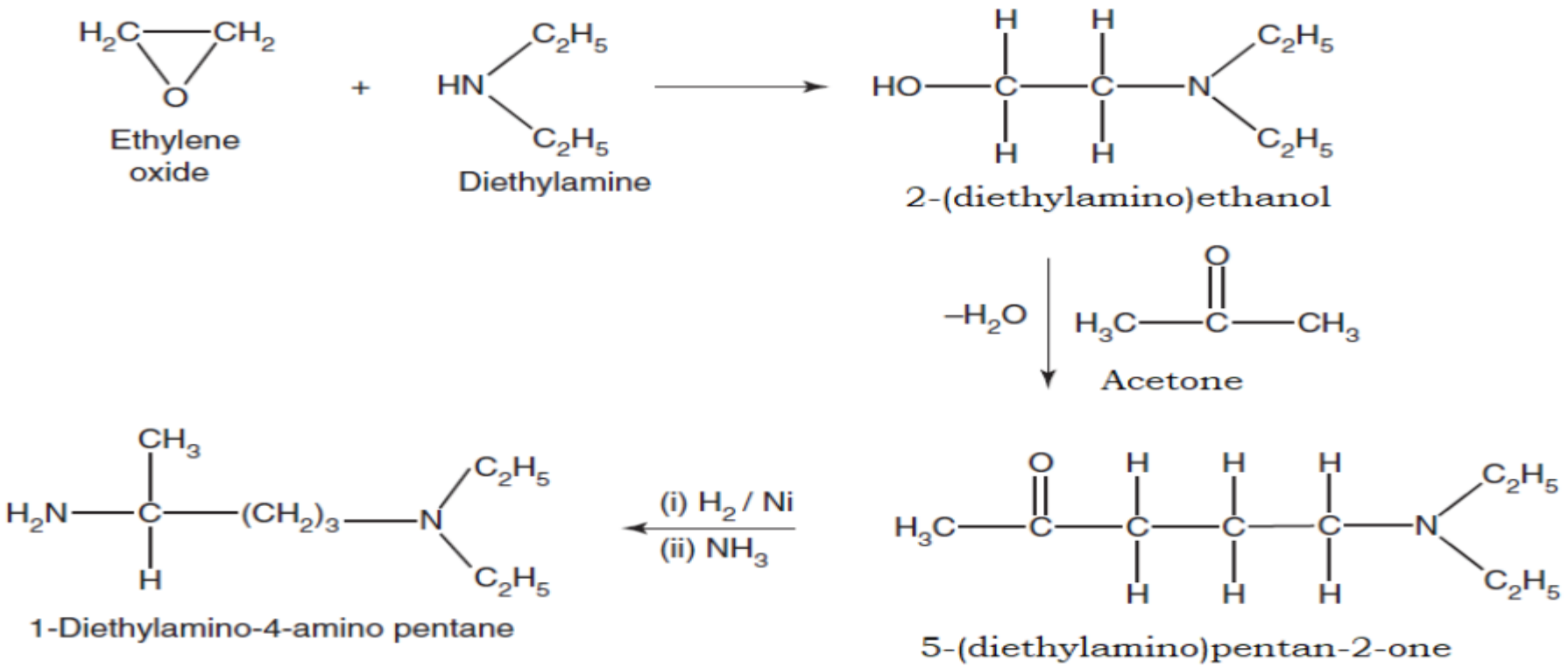
Synthesis of Chloroquine

Step I: Synthesis of 4,7-dichloro quinoline.

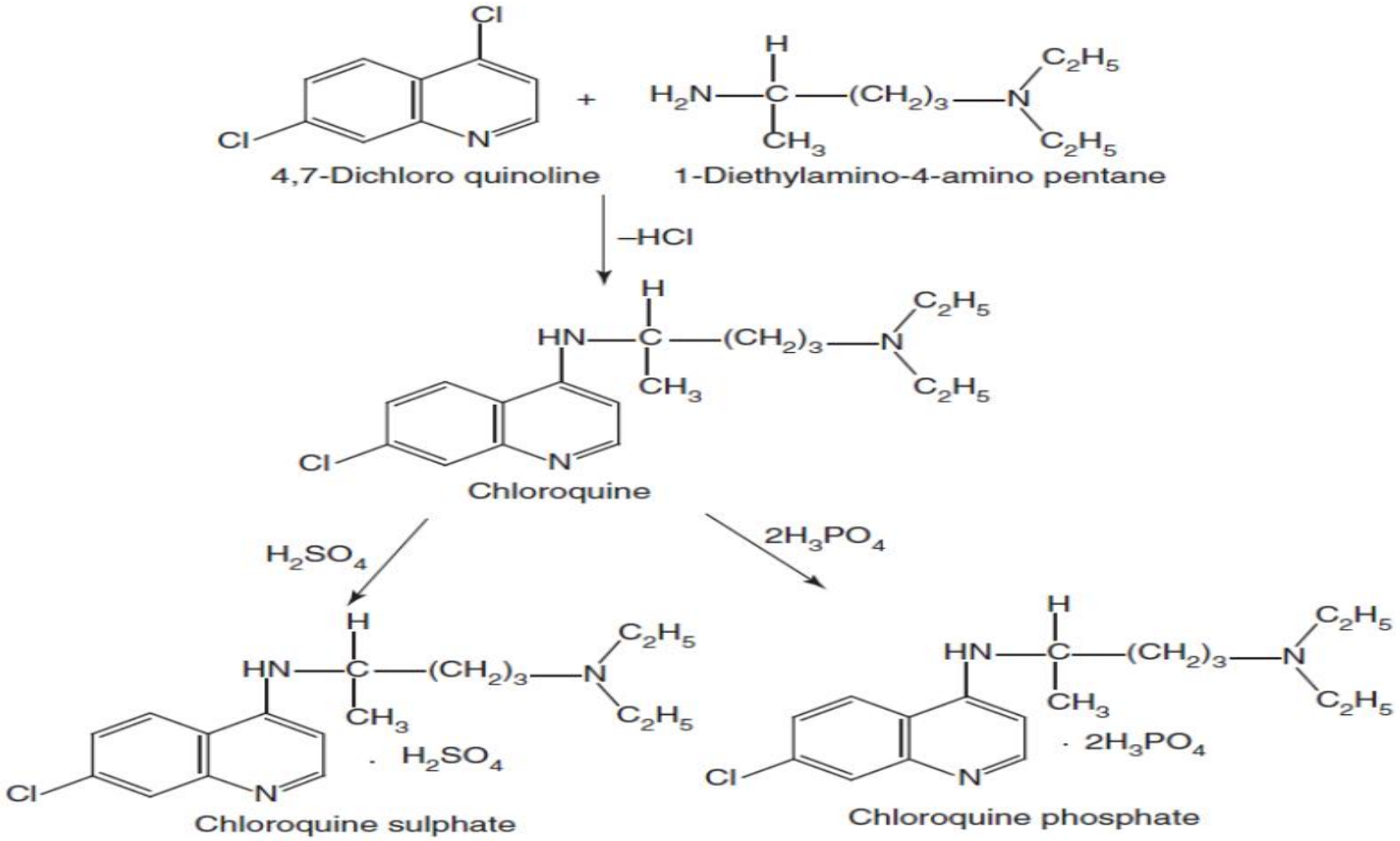



Δ
–CO2
OH
Cl
N
POCl3
Cl
Cl
N
7-chloroquinolin-4-ol
4,7-Dichloro quinoline

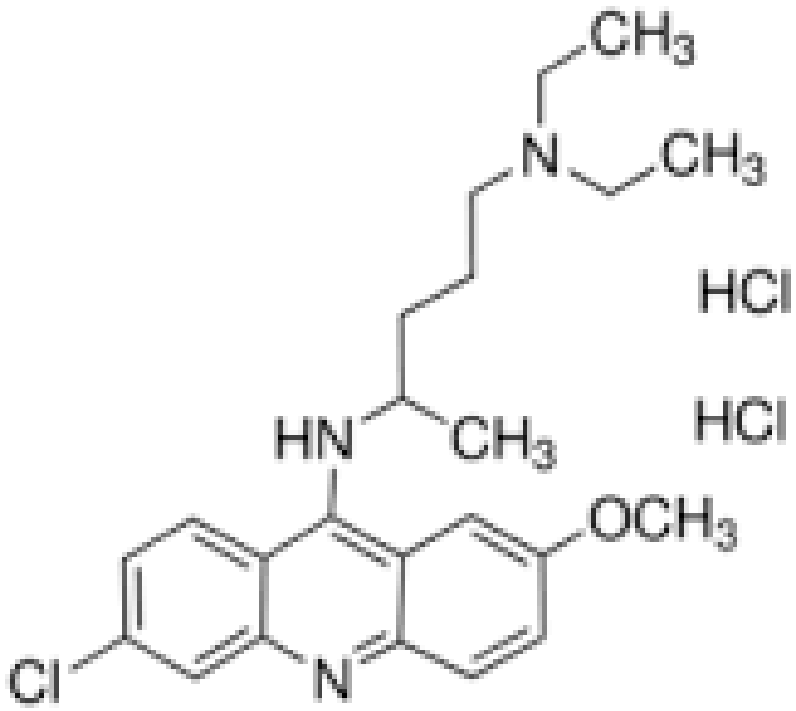
Step-II: *Preparation of 1-diethyl amino-4-amino pentane*



Step III: *Condensation of the products of Step I and Step II*



Quinacrine Hydrochloride



MOA of Quinacrine Hydrochloride

Drug administration

↓

Drug enters infected red blood cells containing parasite

↓

Accumulation of quinacrine inside parasite

↓

Acridine nucleus **intercalates between parasite DNA base pairs**

↓

Inhibition of **DNA replication and RNA transcription**

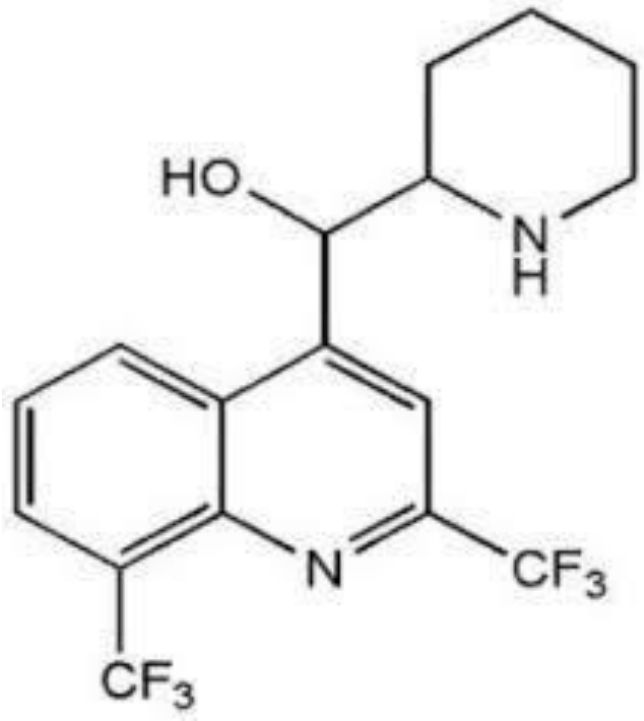
↓

Disruption of nucleic acid synthesis

↓

Death of malaria parasite causing Malaria

Mefloquine



MOA of Mefloquine

Drug administration

↓

Mefloquine enters infected red blood cells

↓

Accumulation in **parasite food vacuole**

↓

Hemoglobin digestion releases **toxic free heme**

↓

Mefloquine **inhibits heme polymerization (heme → hemozoin)**

↓

Accumulation of toxic heme in parasite

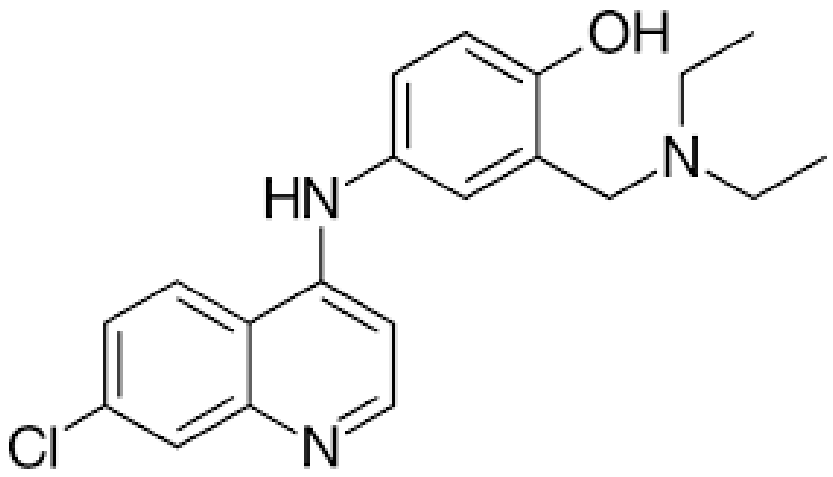
↓

Membrane damage and oxidative stress

↓

Death of malaria parasite

Amodiaquine



Mechanism of Amodiaquine

Drug administration

↓

Drug accumulates in **acidic food vacuole of parasite**

↓

Parasite digests hemoglobin for nutrition

↓

Release of **toxic free heme(Toxic)**

↓

Normally parasite converts **heme → hemozoin (non-toxic)**

↓

Amodiaquine **inhibits heme polymerization**

↓

Accumulation of toxic free heme inside parasite

↓

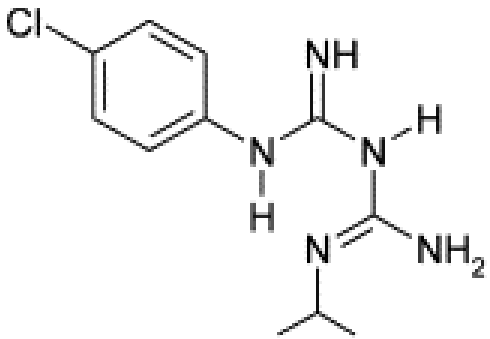
Membrane damage and oxidative stress

↓

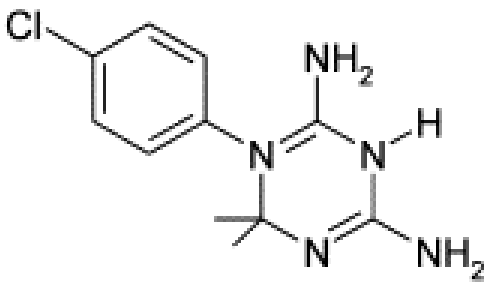
Death of malaria parasite

- **Biguanides and dihydro triazines**

Proguanil and Cycloguanil



proguanil



cycloguanil

Mechanism of Action

Drug administration

↓

Proguanil enters bloodstream and infected liver cells

↓

Drug undergoes **metabolic activation in liver**

↓

Conversion to active metabolite **cycloguanil**

↓

Cycloguanil **inhibits dihydrofolate reductase (DHFR) enzyme** of parasite

↓

Inhibition of **tetrahydrofolate formation**

↓

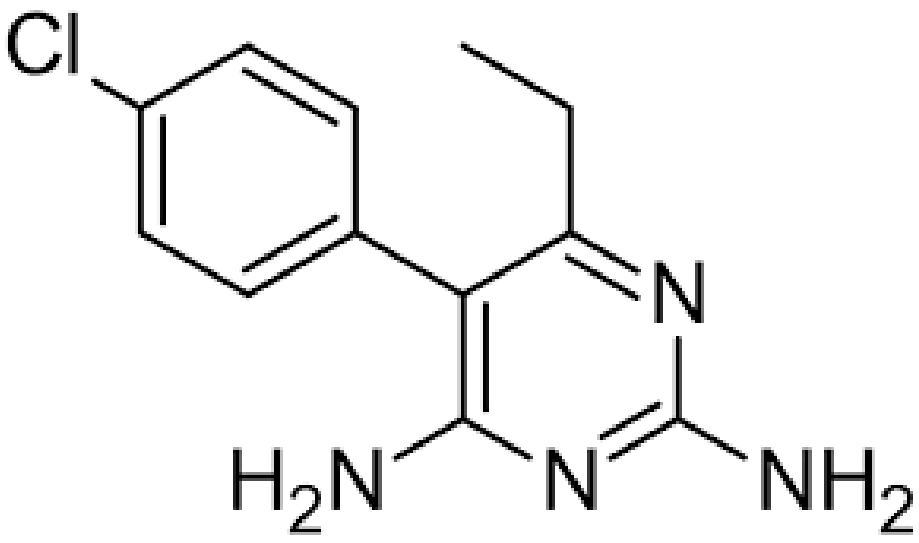
Blockage of **DNA synthesis and cell division**

↓

Death of malaria parasite

- **Miscellaneous**

Pyrimethamine



1. **Pyrimethamine**

Drug administration

↓

Pyrimethamine enters infected red blood cells

↓

Drug reaches malaria parasite

↓

Selective inhibition of **dihydrofolate reductase (DHFR)**

↓

Inhibition of **tetrahydrofolate synthesis**

↓

Reduced synthesis of **purines**

↓

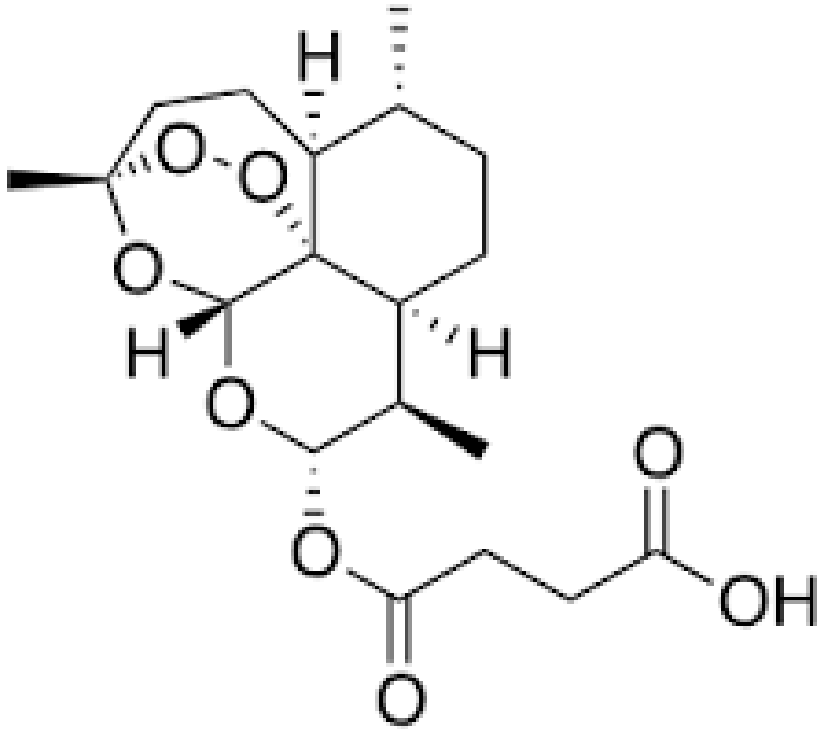
Inhibition of **DNA synthesis and parasite replication**

↓

Death of malaria parasite

- **Miscellaneous**

Artesunate



2. Artesunate

Drug administration

↓

Artesunate enters infected red blood cells

↓

Drug interacts with **heme iron inside parasite**

↓

Cleavage of **endoperoxide bridge** in drug molecule

↓

Generation of **free radicals**

↓

Free radicals damage **parasite proteins and membranes**

↓

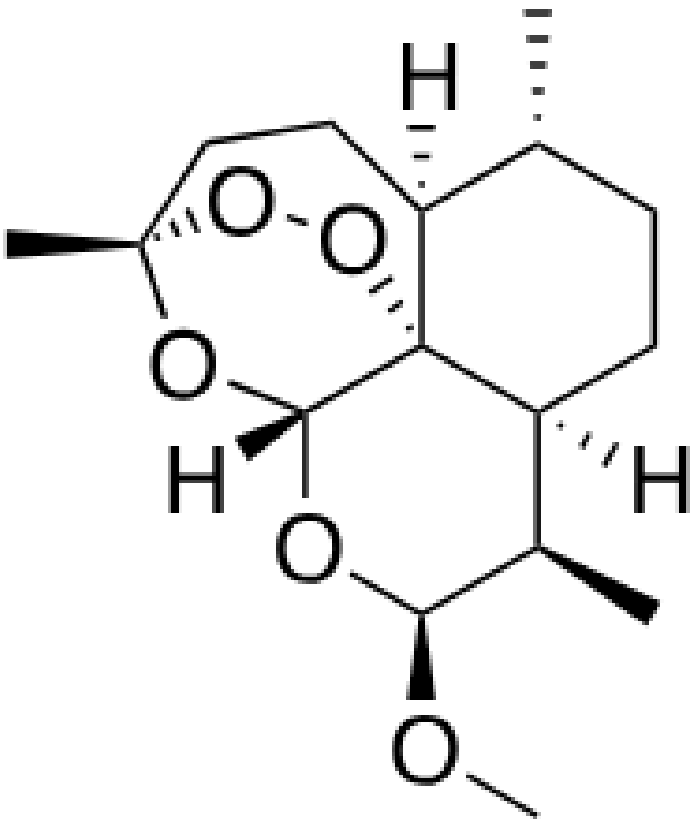
Disruption of parasite metabolism

↓

Rapid death of malaria parasite

- **Miscellaneous**

Artemether

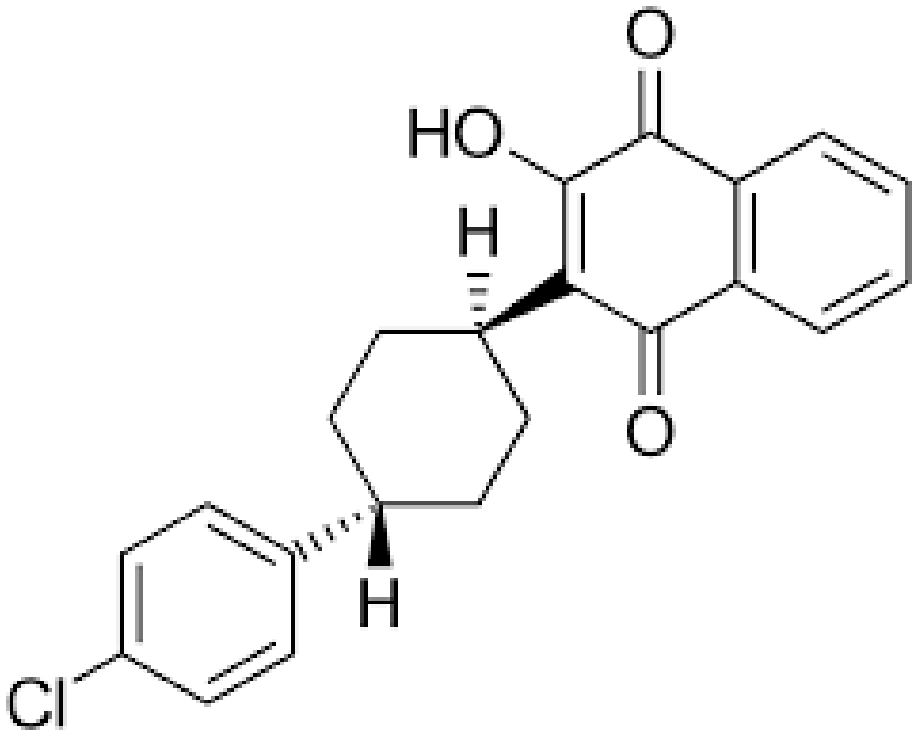


3. Artemether

Drug administration
↓
Artemether enters infected RBCs
↓
Interaction with **heme iron inside parasite**
↓
Breakage of **endoperoxide bond**
↓
Formation of **reactive free radicals**
↓
Damage to parasite proteins and membranes
↓
Disruption of parasite metabolic pathways
↓
Death of malaria parasite

- **Miscellaneous**

Atovaquone



4. Atovaquone

Drug administration

↓

Atovaquone enters infected red blood cells

↓

Drug accumulates in **parasite mitochondria**

↓

Inhibition of **mitochondrial electron transport chain (cytochrome bc1 complex)**

↓

Collapse of **mitochondrial membrane potential**

↓

Inhibition of **ATP synthesis**

↓

Failure of parasite energy metabolism

↓

Death of malaria parasite

THANK YOU